



Vijaybhoomi School of Science and Technology

MID TERM EXAMINATIONS AUGUST 2026

Course Name: MLOps

Course Code: STDE 301

Max. Marks: 20

Problem Statement

A client wants you to develop a machine learning system capable of predicting customer churn based on customer demographics, account information, service usage, and billing details. The system should enable the business team to:

- Identify customers who are likely to leave the service.
- Prioritize retention efforts on high-risk customers.
- Monitor model performance over time.
- Reproduce training results consistently.
- Maintain a reliable and scalable machine learning workflow.

You have been assigned the role of MLOps Engineer. Your responsibility is to design and implement a complete MLOps pipeline that ensures:

- Reproducibility
- Version control
- Experiment tracking
- Automated testing
- Continuous integration

The dataset is available at the following link,

<https://www.kaggle.com/datasets/muhammadshahidazeem/customer-churn-dataset>

The solution should:

1. Ingest and preprocess customer data.
2. Train and evaluate multiple machine learning models for churn prediction.
3. Track datasets and data changes using DVC.
4. Track experiments, parameters, metrics, and model artifacts using MLflow.
5. Implement automated unit tests using Pytest to validate data processing and model functionality.
6. Use Git for version control and collaborative development.
7. Configure GitHub Actions to automatically execute tests whenever code is pushed or a pull request is created.
8. Ensure that the complete workflow can be reproduced by another developer using the project repository.